SOTONSHIP 2026 · TRACK BRIEF

Responsible AI

Friday 16 – Sunday 18 October 2026 · Barclays Eagle Labs Southampton

The challenge

Build a product that makes AI useful and trustworthy for a specific person, and pitch it as a business. You can create a tool that helps others develop, test or govern AI, or an AI-powered product in any sector where responsible design is central to the value it delivers. There is no fixed problem, sector or technology: you choose what to build and show why it matters.

RAi UK brings together technical and wider social perspectives on responsible AI, with research spanning sectors including health, law enforcement and financial services. [1] Your challenge is to turn a specific need into a working product, with evidence of benefit, a clear buyer and a credible way to grow.

SotonShip is about shipping. A focused product that one customer could trial after the weekend is stronger than a broad promise to make all AI safe. Responsibility should change what your product does, how it is tested or what its users can control.

Before you start building, describe your project in one sentence:

> We help **[a specific person]** to **[do or decide something]** using **[AI or an AI assurance tool]**, while reducing **[a specific risk to people]**.

For example: *We help customer-support teams test whether their AI answers are supported by approved sources before a chatbot update reaches customers.*

Who you build for and what fits the track

Choose one main user and one potential buyer; they may be different people. Your user might be a developer, a service worker, a teacher, an AI procurement team, a small business or someone affected by an AI decision. Design around one workflow and identify who benefits, who could be harmed and who can put mistakes right.

Any sector fits. Possible themes include fairness, privacy, security, reliability, transparency, accessibility, human oversight, participation and environmental impact. You do not have to address every theme: choose the risks that matter for your user and make the improvement testable.

The following do not fit, because they have no clear product or cannot be tested responsibly in a weekend:

- a generic chatbot, ethics checklist or dashboard with no specific user, decision or action;
- claims that a system is “fair”, “safe” or “compliant” without a defined scope and supporting evidence;

- an earlier project resubmitted as new work, or a demo that depends on restricted systems or real sensitive personal data.

Each team enters one track only. If your idea also fits FinTech or Southampton City, choose the track where your user and business case are strongest. Here, responsible AI must be central to the problem and the product.

What every project must show

A real problem and people affected. Say who has the problem, how often it occurs and what they do today. Support it with research, published evidence or a conversation with someone affected. Explain a concrete benefit or harm, not just a general concern about AI.

AI at the core. Show the AI capability your product uses or improves. An assurance tool can qualify by testing or governing a running AI system; it does not need an LLM added to its checker. Compare the important step against a simple baseline, such as rules, manual review or the same model without your safeguard.

A product that works. Let a user complete the main task from start to finish. Use public or openly licensed material and reproducible test cases, with sources and dates visible. Label synthetic inputs clearly. Keep test cases separate from examples used to tune the system, and report sample sizes as well as results.

A business case. Pitch a business, not only a research idea: identify the user, paying customer or funder, problem cost, proposed price or funding model, and route to a first pilot. Estimate AI, hosting and human-review costs per use. Explain why a buyer would choose you over existing tools and what changes for a second customer.

Evidence of responsibility and honest limits. Show one measurable improvement and its trade-off, such as fewer unsupported answers at the cost of more handoffs. Demonstrate one failure and what happens next. State who can review, override or challenge the output, and what evidence is still needed before real deployment.

Directions we want to see

These are optional, research-informed starting points. Build one, combine them or ignore them completely. Original ideas are judged on the same terms. The papers support the underlying methods or concerns; the products, buyers and business models below are proposals to validate, not proven commercial outcomes.

A release check for customer-support AI.

RAGAs proposes ways to evaluate retrieval-augmented answers, including their faithfulness to retrieved material. [2] Build a tool that runs test questions against a support chatbot, highlights unsupported claims and creates a review queue. A software company or support agency could pay per bot or test run. Start with licensed public help pages and deliberately answerable and unanswerable questions. Compare against the same bot without the check; measure missed errors and false alarms against human-labelled answers. Automated evaluation is evidence for a reviewer, not proof that every answer is true.

A fairness check before an AI update.

Gender Shades found substantial differences in error rates across gender and skin-type groups in three commercial gender classifiers. [3] That motivates checking performance across relevant groups, rather than relying on one average; it does not establish the same disparities in every domain. Build a release-review tool for one classifier, such as routing customer requests. A product team could buy a subscription covering its models. Use labelled synthetic cases or a suitably licensed benchmark, compare versions and show subgroup error rates, sample sizes and trade-offs. Do not infer sensitive traits from names or faces. Synthetic results demonstrate the workflow, not fairness in a real population.

An evidence assistant for buying AI.

Model Cards and Datasheets for Datasets propose structured documentation of intended uses, evaluations and data provenance. [4] [5] Build an assistant that compares a buyer's needs with supplier documentation, quotes supporting passages and flags missing evidence for a person to review. A small company, procurement team or consultancy could pay per supplier assessment. Test on public model cards plus clearly labelled fictional supplier packs with planted omissions. Compare against a manual checklist; measure missed gaps, invented claims and review time. Sell faster evidence review, not automatic certification or legal approval.

A service tool that knows when to hand over.

SelectiveNet studies prediction with a reject option and the trade-off between error and coverage. [6] Apply that principle to a narrow service workflow: an AI tool routes maintenance requests when evidence is sufficient and sends uncertain cases to a worker. A facilities provider could pay per site or request. Use synthetic requests with known categories, including ambiguous and unfamiliar examples. Compare with routing every request automatically; report errors, the fraction handled automatically and the human queue. This is an adaptation to test, not a claim that the paper validates your workflow. Show what happens when the reviewer is unavailable.

Data

There is no compulsory dataset, model or framework. Choose sources that fit your problem and check reuse rights before building. The resources below include tools and documentation as well as test material; none provides a complete validation of your product.

Source	What it offers	Access and limits
Ragas repository	Evaluation tools for language-model applications; linked to the RAGAs paper [2].	Apache-2.0 code. Model/API use may cost money; validate automated scores with human checks.
Fairlearn	Metrics and methods for assessing model performance across groups.	MIT code. Dataset licences are separate; no single metric guarantees fairness.

Source	What it offers	Access and limits
Hugging Face model cards	Public documentation of intended use, evaluation and limitations.	Check each repository's terms. Public access alone does not grant reuse rights.
Public manuals and help pages	Source material for grounded answers and evidence-review tasks.	Choose content with permission to reuse; record owner, version, URL and date.
Labelled synthetic test cases	Fictional records, planted errors, ambiguous inputs and controlled group comparisons.	State how cases and labels were made. Keep a held-out test set; do not claim population-level validity.

Some services require an account or API key. Keep secrets out of your repository. Record model and tool versions, data licences, test-set construction and the limits of your evidence in the README.

Track rules

The event-wide rules apply: teams of four, hacking runs from Friday 20:00 to Sunday 13:00, and you own what you build. In addition, for this track:

- Build your entry during the event and create its repository after hacking opens. Open-source libraries, templates and pre-trained models are fine, but your own earlier projects are not.
- Credit research, code, models and data, with dates and licences. Disclose AI-generated work, prior components and outside help in your README.
- Label cached and synthetic data. Show uncertainty and avoid safety, fairness or compliance claims beyond what your tests support.
- Do not use private customer records or sensitive personal data in the demo, and do not send confidential information to external AI services. Test only systems you own or are authorised to test.
- Use a sandbox, with no real consequential decisions about people. For health, safety, legal or financial uses, show qualified human review and link to official guidance.

Deliverables and pitch

Each team submits three things on its team page on the SotonShip Live website, before submissions lock at 13:00 on Sunday:

- **A demo video of 1 to 3 minutes** showing your product in use. Every team must upload its video to YouTube and submit the link.
- **A short summary of about 150 words** that explains who your user is, what problem they have and why it matters, what your solution does and how it could scale.
- **A link to your GitHub repository.**

On Sunday afternoon you then pitch in VC for a Day. The **Judging & how to win** page covers the submission requirements and the suggested pitch structure in full.

Judging and how to win

The five criteria and weights are the same as the other tracks. Responsible AI shapes the evidence you present across all five; it is not confined to the final criterion.

Criterion	Weight	What judges should see
Problem and user	20%	A specific need, credible evidence and an understanding of affected people.
AI at the core	20%	A demonstrated AI capability or assurance function, tested against a sensible baseline.
Working product	25%	An end-to-end task, reproducible tests and a demonstrated failure or handoff.
Business and scale	20%	A buyer, value proposition, realistic costs, differentiation and a second-customer plan.
Responsibility and trust	15%	Measured risk reduction, honest trade-offs, provenance and meaningful human control.

On Sunday afternoon every team pitches in VC for a Day, where the audience and judges invest virtual money in the teams they believe in. The top three teams in each track go to a final panel that decides the places. Read the **Judging & how to win** page on the SotonShip Live website for the full process, submission requirements and suggested pitch structure.

For this track, make your pitch answer three questions: what problem and AI-related risk have you evidenced; who would adopt and pay for the product, and why; and what would it take to serve a second customer responsibly? Show the working core, one failure, the measured benefit and the costs of your safeguard. Submission locks at 13:00 on Sunday.

Key times

When	What happens
Friday 16 October, afternoon and evening	Doors open, talks, track introductions and a visit and Q&A with the Lord Mayor of Southampton; hacking opens at 20:00.
Saturday 17 October	Hacking all day; the venue stays open on Friday and Saturday nights, so you can work and stay overnight.
Sunday 18 October, 13:00	Hacking stops and submissions lock, with no extensions.
Sunday afternoon	VC for a Day pitches, the final panel and awards.

The full timetable is on the **Schedule** page. All event times are local to Southampton (BST).

Research and references

The numbered citations and resource names are clickable. Sources checked 15 October 2026. Research examples inform the challenge; they do not imply endorsement of a team or its product.

[1] RAi UK. Research overview.

[2] Es, S., James, J., Espinosa Anke, L. & Schockaert, S. (2024). RAGAs: Automated Evaluation of Retrieval Augmented Generation. EACL System Demonstrations, 150–158.

[3] Buolamwini, J. & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. PMLR 81, 77–91.

[4] Mitchell, M. et al. (2019). Model Cards for Model Reporting. FAT*. arXiv:1810.03993.

[5] Gebru, T. et al. (2021). Datasheets for Datasets. Communications of the ACM. arXiv:1803.09010.

[6] Geifman, Y. & El-Yaniv, R. (2019). SelectiveNet: A Deep Neural Network with an Integrated Reject Option. PMLR 97, 2151–2159.